

Volatility Modeling

The objective of this project is to integrate econometric modeling techniques with modern machine-learning tools to study volatility dynamics in the energy sector and their implications for trading strategies. In particular, the project focuses on forecasting energy-sector volatility and assessing whether improved volatility forecasts can be translated into superior trading and risk-management performance.

The analysis is organized around two main components:

1. Feature Construction
2. Volatility Forecasting

Feature construction provides a set of exogenous signals derived from financial, economic, market, and weather-related variables, while volatility forecasting translates these signals into one-step-ahead predictions of energy-sector volatility. This forecasting step forms the central link between external information and actionable market decisions, serving as the input to downstream trading and risk-management strategies.

1 Volatility Proxy

We begin by defining the volatility measure used throughout the analysis. Volatility is constructed directly from high-frequency returns of the energy-sector ETF (XLE) without decomposing returns into market and idiosyncratic components.

Let $\{P_t^{XLE}\}$ denote the price series of XLE. Scaled, Daily log returns are computed as

$$r_t^{XLE} = \log\left(\frac{P_t^{XLE}}{P_{t-1}^{XLE}}\right) * 100$$

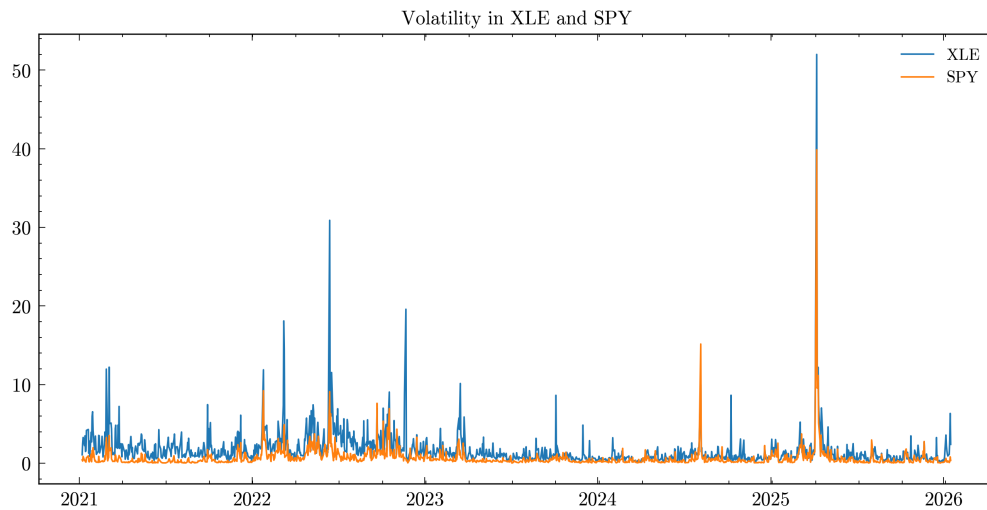
which ensures time additivity and symmetry of positive and negative price movements.

Since latent volatility is unobserved, we proxy daily volatility using realized volatility constructed from intraday return data. Realized volatility has been shown to provide a consistent and informative measure of integrated variance when high-frequency observations are available [2, 3].

Let $\{r_{t,i}^{XLE}\}_{i=1}^n$ denote intraday log returns of XLE on day t , computed over non-overlapping 15-minute intervals, where n is the number of intraday bars within a trading day. Daily realized volatility is defined as

$$RV_t = \sum_{i=1}^n \left(r_{t,i}^{XLE}\right)^2.$$

This realized volatility measure serves as the target variable for all volatility forecasting models considered in the paper, including EWMA, HAR-RV, and GARCH-based specifications.



2 Evaluating Performance

To assess the predictive performance of our volatility models, we adopt an expanding-window forecasting scheme combined with a one-step-ahead prediction structure [1]. Let $\{RV_t\}_{t=1}^T$ denote the realized volatility series, where RV_t is the target variable of interest.

We partition the sample into three components: a training set, a validation set, and a final test set. The training set is used to estimate model parameters, the validation set is employed for model comparison, hyperparameter tuning, and selection of relevant exogenous features, and the test set is reserved strictly for out-of-sample performance evaluation.

Let $\mathcal{F}_t$ denote the information set available at time t , consisting of all past returns and lagged exogenous variables. Specifically,

$$\mathcal{F}_t = \sigma(\{r_s\}_{s \leq t} \cup \{\mathbf{x}_{s-1}\}_{s \leq t}),$$

so that returns at time t depend only on exogenous covariates observed at time $t-1$. All forecasts are generated conditionally on $\mathcal{F}_t$, ensuring that model estimation, tuning, and evaluation respect the information constraints faced by an investor in real time.

Expanding Window Procedure. Suppose the total sample has T observations, and the initial estimation window contains T_0 observations (e.g., $T_0 = 80$ if $T = 100$). The expanding-window method proceeds as follows:

1. Estimate the model using observations 1 through T_0 .
2. Generate a one-step-ahead forecast given the information $\hat{\sigma}_{t+1|t} = \sqrt{\mathbb{E}[\varphi_{t+1}^2 | \mathcal{F}_t]}$.
3. Expand the estimation window to include the new observation, i.e., re-estimate the model using observations 1 through $T_0 + 1$.
4. Produce the next one-step-ahead forecast $\hat{\sigma}_{t+1|t}$.

This process continues until the end of the sample, yielding a sequence of out-of-sample one-step-ahead forecasts $\{\hat{\sigma}_t\}_{t=T_0+1}^T$.

Forecast Evaluation via QLIKE. To evaluate forecast accuracy, we employ the QLIKE loss function, a standard metric in the volatility-forecasting literature between the forecasted and actual volatility measures.

$$\text{QLIKE}(RV_t, \hat{\sigma}_t) = \frac{RV_t}{\hat{\sigma}_t} - \log\left(\frac{RV_t}{\hat{\sigma}_t}\right) - 1.$$

Given the sequence of forecasts, the overall out-of-sample loss is computed as

$$\text{QLIKE}_{\text{OOS}} = \frac{1}{T - T_0} \sum_{t=T_0+1}^T \text{QLIKE}(RV_t, \hat{\sigma}_t).$$

This metric is robust to measurement error in realized volatility and is widely used in comparing volatility models. The model with the lowest average QLIKE loss is considered the best-performing specification.

Volatility Models with Exogenous Proxies

Financial asset returns exhibit pronounced time variation in volatility, characterized by clustering of high- and low-volatility periods. To model this behavior, we focus on forecasting conditional variance directly, rather than specifying a full joint model for returns and volatility. Our framework combines benchmark smoothing methods, realized-volatility regressions, and parametric conditional variance models.

Let r_t denote daily returns and let RV_t denote a realized variance measure constructed from intraday data. We consider three classes of volatility models: EWMA, HAR-RV, and GARCH.

EWMA Benchmark

As a baseline, we employ an exponentially weighted moving average (EWMA) model, which updates conditional variance as

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2,$$

where $\lambda \in (0, 1)$ controls the rate at which past shocks decay. EWMA provides a simple and robust benchmark that captures volatility clustering through adaptive smoothing.

HAR-RV Model

To exploit information contained in realized volatility, we use the heterogeneous autoregressive realized volatility (HAR-RV) model [Corsi2009]. The model captures persistence at multiple horizons by regressing current realized volatility on lagged daily, weekly, and monthly components:

$$\log RV_t = \beta_0 + \beta_d \log RV_{t-1} + \beta_w \log \overline{RV}_{t-1:t-5} + \beta_m \log \overline{RV}_{t-1:t-22} + \varepsilon_t.$$

This structure approximates long-memory behavior in volatility while remaining linear and easy to estimate.

GARCH Model

For comparison with parametric conditional variance models, we estimate a standard GARCH(1, 1) specification,

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2,$$

where ε_t denotes return innovations. The GARCH model provides a widely used reference point for volatility dynamics driven by past shocks and persistence.

Exogenous Proxy Variables

Rather than introducing exogenous variables directly into the return or conditional variance equations, we incorporate external information through a two-stage volatility modeling strategy based on economically motivated proxy variables. These proxies include macro-financial indicators (e.g. interest rates and credit spreads), commodity and sector ETFs, and weather-based anomaly measures relevant to energy markets.

In the first stage, baseline volatility forecasts are generated using EWMA, HAR-RV, or GARCH models. In the case of GARCH, the model produces a sequence of conditional variance estimates $\{\hat{\sigma}_t^2\}$ based solely on past returns and volatility dynamics.

In the second stage, the baseline volatility forecast is treated as an input to a regression that conditions volatility on exogenous information. Specifically, we estimate regression models of the form

$$\log \hat{\sigma}_{t+1}^2 = \gamma_0 + \boldsymbol{\gamma}^\top \mathbf{x}_t + u_{t+1},$$

where $\mathbf{x}_t$ denotes a vector of proxy variables observed at time t . This formulation allows exogenous signals to adjust the level of predicted volatility without altering the internal dynamics of the underlying GARCH process.

An analogous conditioning step is applied to HAR-RV forecasts, where proxy variables are included as additional regressors in the realized-volatility regression or used to post-process predicted volatility levels through ranking and regime classification. This design preserves the stability and interpretability of the baseline volatility models while allowing forecasts to respond flexibly to observable economic and environmental conditions.

Across all specifications, the resulting one-step-ahead volatility forecasts $\widehat{RV}_{t+1}$ are evaluated using the QLIKE loss, ensuring comparability between baseline and proxy-augmented models.

3 Feature Importance and Incorporation of Weather Data

A key objective of this project is to determine whether weather anomalies contain incremental predictive power for realized volatility in the energy sector. To evaluate the usefulness of potential predictors, we adopt a greedy feature-selection procedure based on out-of-sample performance measured by the QLIKE loss.

Let $\mathcal{X}$ denote the set of candidate exogenous features, which includes various measures of weather anomalies (e.g., temperature deviations, precipitation shocks, wind anomalies, and regional extreme-weather indicators), as well as other macro econometric features (fed funds, natural gas rate, VIX). Each feature is lagged appropriately so that only information available at time t is used when forecasting RV_{t+1} .

Greedy Feature Selection Procedure. We employ the following iterative scheme, using GARCH(1, 1) as the baseline specification:

1. Estimate a standard GARCH(1, 1) model using only past returns, and compute its out-of-sample QLIKE loss on the validation set. Denote this loss by L_{base} .
2. For each candidate feature $x \in \mathcal{X}$, augment the GARCH model with x_t as an exogenous regressor in the variance equation, re-estimate the model, and compute the corresponding out-of-sample QLIKE loss $L(x)$.

3. Compute the improvement in forecast accuracy

$$\Delta(x) = L_{\text{base}} - L(x),$$

and rank features according to the magnitude of $\Delta(x)$.

4. If desired, the procedure may be repeated by adding the most informative feature to the model and evaluating the marginal contribution of the remaining features.

In this framework, weather variables enter as exogenous predictors in the variance equation. By evaluating their incremental reduction in QLIKE, we can assess whether extreme temperature deviations, storm activity, or other meteorological shocks have measurable effects on energy-sector volatility. Features that consistently reduce the QLIKE loss are retained for use in the final forecasting models (including XGBoost), while features with negligible or negative contributions are excluded.

This approach enables a transparent, data-driven method for identifying which weather indicators meaningfully enhance volatility forecasts.

4 Results

Before estimating models with time-varying volatility, we first assess whether the return series exhibits conditional heteroskedasticity. If volatility were constant over time, a GARCH-type specification would be unnecessary. To formally evaluate the presence of volatility clustering, we apply Engle's ARCH test to the return series.

The ARCH test evaluates whether the squared residuals from the mean equation exhibit serial correlation, which would indicate time-varying conditional variance. Under the null hypothesis of homoskedasticity, lagged squared residuals should not have explanatory power for current squared residuals. The test yields a p-value of 0.03, which is below the 0.05 significance level. Accordingly, we reject the null hypothesis and find statistical evidence of conditional heteroskedasticity, supporting the use of GARCH-type volatility models.

Feature Importance and Redundancy Control

To assess the contribution of individual exogenous variables, we evaluate feature importance through their incremental impact on out-of-sample forecast accuracy. Starting from a baseline HAR-RV specification, exogenous features are added sequentially and model performance is evaluated using the QLIKE loss. Lower QLIKE values indicate improved volatility forecasts.

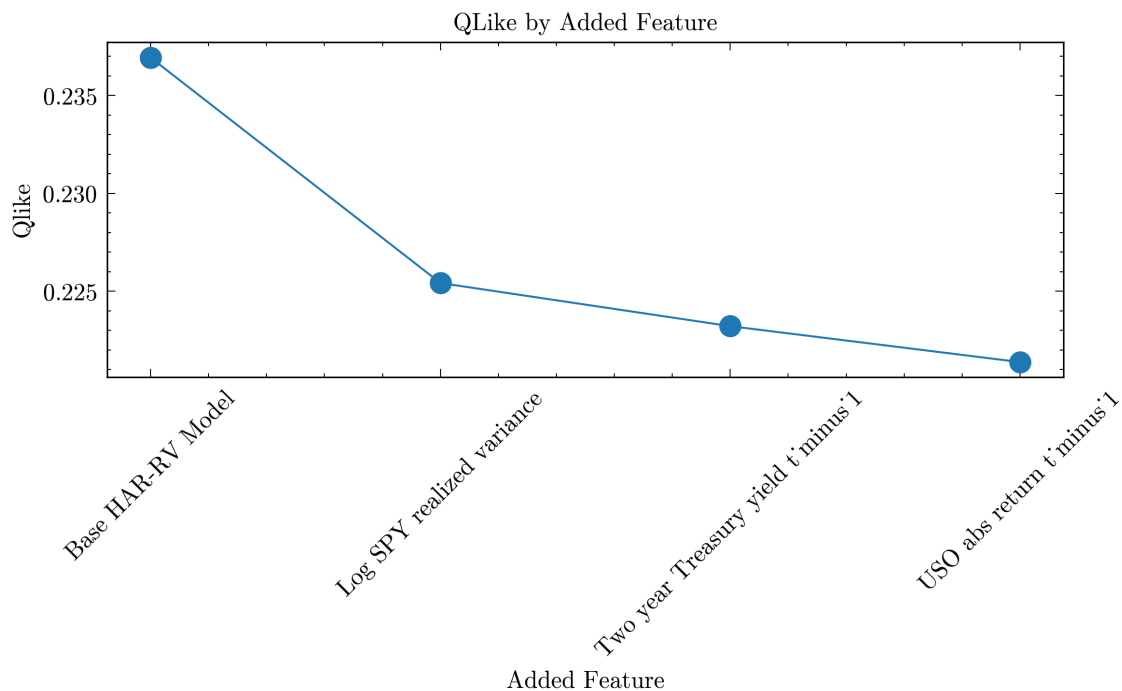


Figure 1: Incremental QLIKE improvement from sequentially adding exogenous features

Figure 1 illustrates the change in QLIKE as additional features are introduced. The inclusion of log realized volatility of the broad market (SPY) produces a substantial reduction in forecast loss relative to the baseline model, indicating

that market-wide volatility contains information relevant for energy-sector volatility. Adding the two-year Treasury yield further improves performance, suggesting that short- to medium-term interest rate conditions contribute incremental explanatory power. Finally, the inclusion of absolute returns on the energy commodity ETF (USO) yields an additional, though smaller, improvement in QLIKE.

Because many candidate features are highly correlated, especially those related to financial markets and volatility, we explicitly control for redundancy. Prior to model estimation, highly correlated variables are identified using pairwise correlation analysis. When two features exhibit strong correlation, the variable with weaker incremental forecast contribution is excluded from the final specification. This procedure reduces multicollinearity, improves parameter stability, and limits overfitting.

The final set of selected exogenous variables consists of log realized SPY volatility, the two-year Treasury yield, and lagged absolute returns of the USO ETF. These features are used consistently across proxy-augmented HAR-RV and GARCH-based models, ensuring comparability across specifications while retaining only the most informative external signals.

In-Sample Model Fit

We examine in-sample model fit relative to realized volatility, smoothed using a 7-day moving average for visualization. Figure 2 compares fitted volatility paths across EWMA, GARCH, HAR-RV, and proxy-augmented specifications.

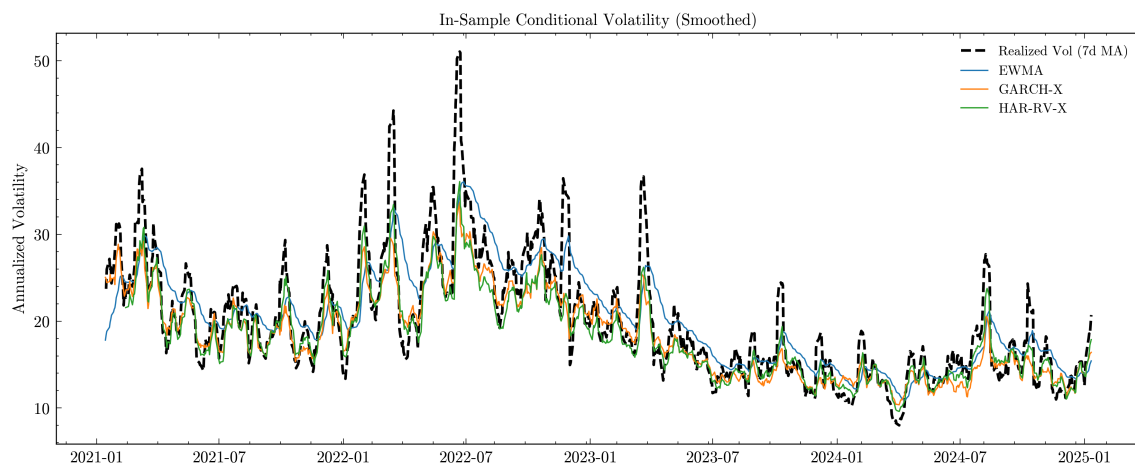


Figure 2: In-sample fitted volatility versus realized volatility (7-day moving average)

All models capture the broad level and persistence of volatility; however, models that incorporate realized volatility components and exogenous proxy information track short-run fluctuations more closely. In particular, HAR-RV-X(3) and GARCHProxyX exhibit improved responsiveness during volatility spikes while avoiding excessive persistence during calm periods.

Out-of-Sample Forecast Performance

We next evaluate one-step-ahead out-of-sample volatility forecasts. Figure 3 plots predicted volatility against realized volatility over the evaluation window. Proxy-augmented models adapt more quickly to changes in volatility regimes and more accurately capture the magnitude of volatility spikes.

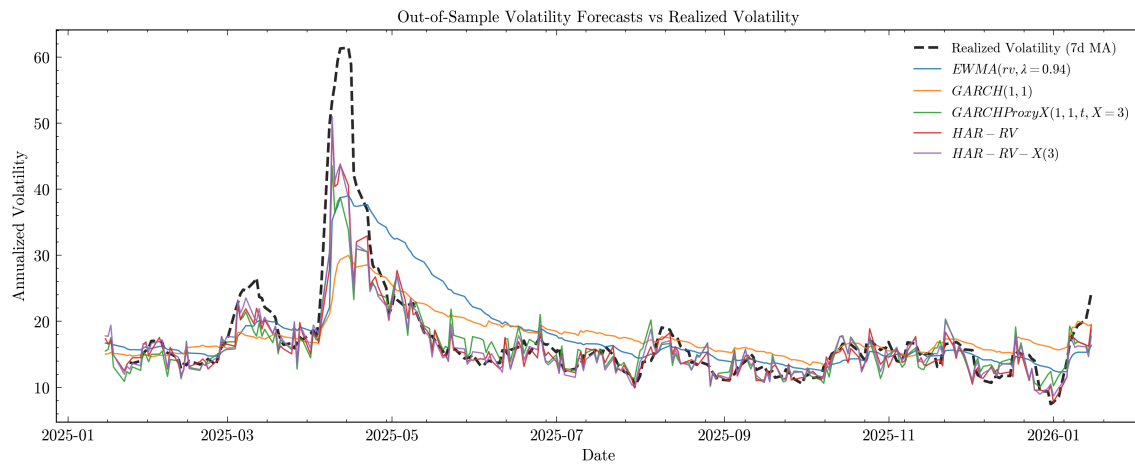


Figure 3: Out-of-sample volatility forecasts versus realized volatility

Forecast accuracy is assessed using the QLIKE loss, root mean squared error (RMSE), and correlation with realized volatility. Table 1 summarizes the results.

Table 1: Out-of-Sample Volatility Forecast Performance

Model	QLIKE	Correlation	RMSE	n
GARCHProxyX(1,1; X=3)	0.2718	0.4853	3.2410	252
HAR-RV-X(3)	0.2749	0.4584	3.2199	252
HAR-RV	0.2940	0.4214	3.2677	252
EWMA (RV, $\lambda = 0.94$)	0.3827	0.2495	3.4562	252
GARCH(1,1)	0.3845	0.2608	3.4563	252

Proxy-augmented models consistently outperform baseline specifications across all evaluation metrics. The GARCH-ProxyX model achieves the lowest QLIKE loss and highest correlation with realized volatility, while HAR-RV-X attains the lowest RMSE.

Residual Diagnostics

To assess overall model fit, we also examine residual dispersion. Table 2 reports residual variance and standard deviation for each model. Consistent with the out-of-sample results, proxy-augmented HAR-RV and GARCH specifications produce the lowest residual variance, indicating a tighter fit to realized volatility.

Table 2: Residual Variance and Standard Deviation Across Volatility Models

Model	Residual Variance	Residual Std. Dev.
HAR-RV-X(3)	10.3104	3.2110
GARCHProxyX(1,1; X=3)	10.4291	3.2294
HAR-RV	10.6116	3.2575
GARCH(1,1)	11.9656	3.4591
EWMA (RV, $\lambda = 0.94$)	11.9925	3.4630

References

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